

Heat up + Calibration

Phase 01 | Operator explanation and control logic

Release	Build	Reviewed	Executable
0.9.41	2026.09.04-r20	2026-09-04	Phase 01

Safety boundary: this guide describes the software sequence only. Follow the current laboratory SOP, confirm the physical chamber and instruments, and remain supervised. Software limits do not replace hardware interlocks or the emergency procedure.

Purpose

Warm the CK-1 evaporator under supervised current control, qualify the deposition signal, and calculate the CK-1-to-sample QMB thickness ratio used by Phase 03.

Scope and ownership

The phase script owns serial I/O, ramp/PID control, QMB quality checks, shutter gating, plotting and safe shutdown. The launcher starts it as a separate process and verifies serial release after exit.

Operator sequence

1	Enter a meaningful sample/run name and verify the displayed phase and recipe.
2	Complete the physical pre-flight and confirm the QMB, Arduino, pyrometer, Keysight and pressure-gauge connections.
3	Start the phase. The output is enabled only after the startup current is verified at 0 A; the first automatic setpoint is 0.005 A.
4	Supervise the warm-up. The default ramp uses 0.005 A steps every 15 s until 100 deg C, then slope/PID control according to the selected validated mode.
5	Open the shutter only when the dashboard prompts and the minimal process gate is satisfied: target CK-1 temperature and averaged CK-1 rate, plus the calibration target condition.
6	Keep the shutter open until the sample-QMB relative thickness reaches 2.0 angstrom and remains at or above target for five continuous seconds of fresh readings.
7	Close the shutter when prompted. The ratio is frozen at stable confirmation so operator reaction time after the endpoint is not included.
8	Use Finish for the normal controlled ramp-down and wait for the launcher to confirm the process and serial handoff are complete.

Packaged defaults

Parameter	Default	Meaning
CK-1 temperature guide	242 deg C	Process guide and temperature-mode target.
CK-1 rate guide	0.40 angstrom/s	Minimum averaged rate for the shutter gate and rate-mode target.
Calibration target	2.0 angstrom	Sample-QMB relative thickness endpoint.
Stable endpoint	5 continuous s	Fresh sample-QMB readings at or above target.
Automatic current cap	0.660 A	Highest editable automatic command.
Software hard stop	0.680 A	Fixed current stop; not an operator recipe value.
CK-1 watchdog	255 deg C	Independent safety stop for CK-1 temperature.
PID temperature band	+/-0.7 deg C	Default temperature control dead band.

Control and safety logic

The selected feedback mode controls how the current request is shaped: temperature mode preserves the established temperature loop; rate mode uses conservative CK-1 QMB rate feedback after warm-up; compound mode uses a rate-informed temperature target while the inner temperature PID remains the current command authority.

QMB plausibility filters and independent watchdogs protect the control path. A stale or implausible signal is not allowed to silently satisfy a readiness condition. The shutter transition is intentionally minimal and visible to the operator; controller-internal slope and trend metrics do not add a hidden shutter timer.

If the phase reports a high temperature, electrical limit, pressure problem, stale instrument signal or other safety fault, use Abort / Safe Stop and inspect the physical chamber before attempting another run.

Finish and abort behavior

Normal completion	Abort / Safe Stop
Finish enters the normal ramp-down, reaches 0 A, switches the Keysight output OFF, saves summaries/plots and exits the phase.	Abort prioritizes the electrical safe state and stops the phase. Do not close the launcher by force; wait for process exit and serial handoff verification.

Data written

A new folder is reserved below Data Samples/Heat up + Calibration Data using the pattern Heat up + Calibration <sample> data NN. The folder contains effective parameters, telemetry, QMB quality events, calibration summaries, shutter markers, plots and final report files as applicable.

Important record	Use
Calibration summary	Stable endpoint values and the frozen CK-1/sample thickness ratio.
CSV/text telemetry	Audit trail for temperatures, currents, thicknesses, rates and safety events.
Plots and snapshots	Visual review of the process and open/close shutter interval.

Before handing over

Confirm the ratio is present in the saved summary, record any physical observations in the laboratory run record, and start Phase 03 only after the launcher displays and confirms the ratio.